

of applications, making software development even more efficient.

The origins: Virtualisation

Virtualisation started back in the 1960s when IBM introduced the idea of virtual machines (VMs). At that time, each computer could only run one task and one operating system (OS) at a time. If a company needed to run several programs, they needed a separate computer for each one, which wasn't efficient. For example, if a business wanted to run both payroll software and inventory management software, they would need two separate machines, even if neither was fully utilising the computer's resources.

Virtualisation changed all that. It allowed several OS instances to run on the same physical machine, helping companies use their hardware more efficiently. IBM's System/360, launched in 1964, was one of the first computers to allow multiple users to share the same machine while keeping their work separate. For instance, a university could use the System/360 to run different programs for both administrative tasks and student research on the same machine without interference. This breakthrough laid the foundation for the modern world of cloud computing.

As the years went by, in the 1980s and 1990s, virtualisation kept improving. Advanced operating systems like UNIX helped create stronger virtual environments. As computers got faster and cheaper, more businesses began using virtualisation, especially with big mainframe computers. This made it easier to run several applications on fewer machines, saving money and making everything simpler to manage.

In the late 1990s and early 2000s, virtualisation saw a major shift with x86 virtualisation. This technology allowed personal computers to run multiple operating systems at the same time. Before this, only large and expensive servers could handle such tasks. But

x86 virtualisation made it possible for even everyday computers with common processors (like Intel and AMD chips) to use this technology, making it more affordable and accessible to businesses and individuals. Companies like VMware developed software that made this process even easier. For instance, VMware's product, VMware Workstation, enabled people to create and manage virtual machines (VMs) on their PCs. This was a game-changer because businesses could now test new applications, set up development environments, and create disaster recovery systems much more easily. For example, an IT department in a large company could use VMware to run both Windows and Linux environments on the same physical server. This allowed them to host their company's website while also running internal payroll software on the same machine, saving both costs and resources.

However, VMs had their drawbacks. Each virtual machine needed its own operating system, which used up a lot of memory and processing power. This made VMs slow to start and resource-heavy. For example, starting a new VM could take several minutes, wasting valuable time. As businesses needed faster deployment times and more flexible applications, the inefficiencies of traditional VMs became more noticeable. Managing a large number of VMs was also tricky. Each one needed separate updates, security patches, and monitoring, which made it hard to manage them. As businesses grew, these limitations became clear, and there was a need for faster, lighter solutions.

Despite these challenges, the evolution of virtualisation from the 1960s to the 2000s was essential and paved the way for the next big development: containerisation.

Containers: The game changer

Containers began to gain attention in the mid-2000s, particularly with the introduction of Linux containers

(LXC) in 2008. However, it wasn't until Docker's launch in 2013 that containers became widely accessible and easy to use, benefiting everyone—from small startups to large enterprises. Docker offered a simple way to package software, including the code, libraries, and dependencies, into one container.

Think of containers like virtual machines, but much lighter and more efficient. Unlike VMs, which need their own operating system, containers share the same OS but keep the applications inside them isolated from each other. This makes containers faster to start; they use less memory, and are easier to move between different environments—whether it's a developer's laptop, a testing server, or even the cloud.

Docker made this process simple. It allows developers to 'containerise' an app, ensuring it runs the same everywhere. For example, an app built on a laptop using Docker can run just as smoothly on a server or in the cloud without any changes. This consistency is one of the main reasons containers have become so popular.

As Docker succeeded, a whole ecosystem of tools was built around containers. One of the most important tools is Kubernetes, originally developed by Google. Kubernetes helps manage containers at scale by automating the process of deploying, scaling, and maintaining them. If a company runs hundreds or thousands of containers, Kubernetes makes it easier to keep everything running smoothly.

One of the more innovative aspects of containers is their support for microservices. In microservices, an app is broken down into smaller, independent parts, each of which can be developed and managed separately. This makes app development faster and more flexible, as developers can quickly update or improve individual parts of an app.

Containers are also key in CI/CD (continuous integration/continuous deployment), a modern software practice that automates building,